

Overview: two sequential stages

Stage 1 (Eligibility Check) must pass before Stage 2 (Search Engine) ever runs. If a parcel fails Stage 1, it is not eligible and proceeds directly to standard RTO handling — it is never scored for matching.

Stage 1: Eligibility Check

Gate a — Seller opted in

Binary check. If the seller has not enrolled in the Rescue program, stop here — the parcel is not eligible, regardless of any other factor.

Gate b — Category tier and base weight

Every item maps to one of three tiers, each carrying a fixed weight:

Tier	Meaning	Base weight
A	Low personalization (home/kitchen, generic fashion)	1
B	Personalized items (clothing sizes, footwear)	0.8
C	Not eligible (very low value, or RTO cost already beats resale value)	0

If the item is Tier C, weight is 0 — this mathematically zeroes out everything downstream, so it functions as a second stop condition.

Output of Stage 1: a single number, the base weight (1, 0.8, or 0), attached to the parcel.

Stage 2: Search Engine — scoring each potential buyer

For every nearby candidate buyer, the system pulls up to three independent signals and scores each. A single candidate may have one, two, or all three signals present — each is looked up separately.

Signal 1 — Live cart / wishlist match

Is this exact or similar product in the candidate's cart or wishlist right now? Score depends on how recently the activity occurred.

Signal type	Time since activity	Score
Cart match	0–3 hrs	1
Cart match	3–24 hrs	0.8
Cart match	24–72 hrs	0.5
Cart match	72+ hrs	0.2
Wishlist match	0–3 hrs	0.9
Wishlist match	3–24 hrs	0.7
Wishlist match	24–72 hrs	0.4
Wishlist match	72+ hrs	0.1

Wishlist scores slightly lower than cart at every time bucket, since a cart add signals stronger purchase intent than a wishlist save.

Signal 2 — Recent search history

Did the candidate search for this product or category without buying?

Time since search	Score
1–3 days	0.6
3–7 days	0.4
>7 days	0.2

Signal 3 — Regional demand pattern

Does this area generally buy this kind of product?

Pattern	Score
High demand in past 30 days	0.5
Multiple past orders (this buyer specifically)	0.3
General category demand only	0.1

If a candidate has more than one signal present, take the **HIGHEST** of the applicable signal scores — do not sum them. Each signal is an independent way of detecting interest; the goal is the strongest evidence, not an inflated total.

Exact vs. similar match — the final multiplier

Once a signal score is chosen, one more multiplier applies, based on whether the matched item is the exact SKU or only similar:

Category tier	Exact SKU match	Similar match
A	1	0.8
B	1	0.4

Tier B drops much further for a similar (non-exact) match than Tier A does — deliberately. A similar sofa cushion is still basically fine; a similar (wrong size) pair of shoes is a real gamble.

The final formula

Final Match Score = Base weight (Stage 1) × Best signal score (Stage 2) × Exact/similar multiplier

Worked example 1

A Tier B item (clothing), seller opted in. Nearest candidate has a cart match from 5 hours ago, exact same SKU.

- Base weight (Tier B) = 0.8
- Best signal score (cart match, 3–24 hr bucket) = 0.8
- Match type (exact, Tier B) = 1

Final Score = $0.8 \times 0.8 \times 1 = 0.64$

Worked example 2

Same item, second candidate: a search-history hit from 2 days ago, similar (not exact) item.

- Base weight = 0.8
- Signal score (search, 1–3 day bucket) = 0.6
- Match type (similar, Tier B) = 0.4

Final Score = $0.8 \times 0.6 \times 0.4 = 0.192$

The system ranks all candidates by this final score and offers the item to the top 3 (or chosen top-N), highest score first.

